	- 2 ; - 3 ;	**(1,0)** **(0,5)**
		3

문제 2. 물이 든 용기 (7점)

1. .

$$p = p_0 + \rho g(L - x_0). \quad (1)$$

,

.

$$p_0 Sh = vRT_0, \quad (2)$$

$$pS(h - x_0) = vRT_0, \quad (3)$$

S , v .

(1) - (3) :

$$\rho g x_0^2 - [p_0 + \rho g(L + h)]x_0 + \rho g h L = 0, \quad (4)$$

$$x_0 = \frac{1}{2}\left[\frac{p_0}{\rho g} + L + h \pm \sqrt{\left(\frac{p_0}{\rho g} + L + h\right)^2 - 4hL}\right]. \quad (5)$$

(5) .

. $x_0 = h$ $p_0 = 0$, $x_0 = 0$ $L = 0$,

$$x_0 = \frac{1}{2}\left[\frac{p_0}{\rho g} + L + h - \sqrt{\left(\frac{p_0}{\rho g} + L + h\right)^2 - 4hL}\right]. \quad (6)$$

$$x_0 = 7.86 \cdot 10^{-2}\, m. \quad (7)$$

2. , x

$$p(x) = p_0 + \rho g(L - x). \quad (8)$$

3. x

$$p(x)S(h - x) = vRT(x), \quad (9)$$

(1)

$$T(x) = T_0\left(1 - \frac{x}{h}\right)\left(1 + \frac{\rho g(L-x)}{p_0}\right). \quad (10)$$

4.

$x = 0$. (10)

$$T_m = T_0\left(1 + \frac{\rho g L}{p_0}\right), \quad (11)$$

$$T_m = 350\, K. \quad (12)$$

5.

$$\Delta U = \frac{5}{2}vR(T - T_0) = \frac{5}{2}\rho g L S h, \quad (13)$$

$$A = \int_0^{x_0} p(x)S dx = \frac{1}{2}p_0 SL\left(1 + \frac{p_0}{2\rho g L} + \frac{\rho g L}{2p_0}\left[1 + \frac{2h}{L} - \frac{h^2}{L^2}\right]\right) -$$

$$-\frac{1}{4}p_0 SL\left(1 + \frac{\rho g(L-h)}{p_0}\right)\sqrt{\left(1 + \frac{h}{L} + \frac{p_0}{\rho g L}\right)^2 - \frac{4h}{L}}. \quad (14)$$

1

$$Q = \Delta U + A, \quad (15)$$

(13) (14)

$$Q = \frac{1}{2}p_0 SL\left(1 + \frac{p_0}{2\rho g L} + \frac{\rho g L}{2p_0}\left[1 + \frac{12h}{L} - \frac{h^2}{L^2}\right]\right) -$$